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When utility meets ethics: A stakeholder perspective on agentic information systems delegation

Kashif Saeed *, Victor R. Prybutok

Department of Information Technology & Decision Sciences, G. Brint Ryan College of Business, University of North Texas, 1155 Union Cir, Denton, TX 76205, United States

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ABSTRACT

Organizations increasingly use agentic information systems (agentic IS) to achieve performance-driven objectives. However, agentic IS raise ethical concerns, such as fairness, bias, autonomy, transparency, responsibility, and privacy. Over 70 % of the stakeholders, such as users, have expressed concerns about the various business applications of agentic IS. However, limited research has examined how stakeholders evaluate utility versus ethics in delegating tasks to agentic IS. Our results show that stakeholders assess agentic IS for delegation using a combination of rules-based and outcome-based ethical thinking, which values both AI utility and ethical considerations. Moreover, when agentic IS embeds ethical capabilities, stakeholders view both agentic IS and delegating tasks to them as ethically sound. Furthermore, stakeholders consistently prefer agentic IS with human oversight over fully autonomous systems. These insights help organizations optimize efficiency through agentic IS while promoting ethical adoption by respecting stakeholder preferences. This research contributes to the IS literature by integrating ethics into the agentic delegation framework, thereby establishing a foundation for future research in the developing field of agentic IS.

1. Introduction

Agentic information systems (agentic IS), more commonly referred to as artificial intelligence (AI) or AI systems, have the intrinsic capability to gather knowledge, adapt to changing conditions, function autonomously, and demonstrate self-awareness without external prompts (Baird & Maruping, 2021; Floreano & Wood, 2015; Russell, 2019). They differ from traditional information systems (IS) in their capacity to facilitate the transfer of rights and responsibilities to and from human agents, a process defined by Baird and Maruping (2021) as *delegation*. This requires a paradigm shift from humans "using" IS to solve a problem or a task, to humans delegating a task to agentic IS (Baird & Maruping, 2021). Baird and Maruping (2021) characterized delegation as a dyadic relationship between humans and agentic IS, referring to the human participant as the "human agent," typically the principal or delegator in most delegation scenarios. However, in accordance with stakeholder theory, we contend that the role transcends from being merely the principal or delegator to encompassing a broader stakeholder perspective that includes all parties affected by the delegation decision.

Organizations adopting agentic IS frequently operate under the assumption that the delegation of a task to agentic IS inherently delegates the associated ethical responsibilities to agentic IS (Garcia, 2024). We contend that when a task is delegated to an agentic IS, the ethical responsibility ultimately resides with the delegator, although it may extend to other parties. Consequently, delegation serves as the fundamental mechanism by which ethical issues related to agentic IS emerge. Thus, the ethics of agentic IS has become an essential topic for academics and practitioners (Akter et al., 2021; Dwivedi et al., 2021).

Agentic IS are designed to adapt to changes to pursue goals embedded in their algorithmic logic (Carroll et al., 2024; Russell & Norvig, 2021), offering utility and performance gains to humans and organizations. In the context of organizational AI delegation, goal-orientation pertains to the formulation and application of agentic IS to attain particular outcomes—usually performance gains, cost reductions, efficiency improvements—that align with the objectives of the organizational delegator. This corresponds to a bounded or agent-relative variant of utilitarianism, wherein utility maximizing is confined to the objectives and values of the decision-making agent (the delegator or the organization), rather than pursuing an impartial maximization for all impacted stakeholders (Mill, 2016; Sen, 1982). These outcomes enhance overall utility within the organizational

* Corresponding author.
E-mail addresses: kashif.saeed@unt.edu (K. Saeed), victor.prybutok@unt.edu (V.R. Prybutok).

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system, which is consistent with instrumental or act utilitarianism (Sen, 1979). However, this focus on performance as the primary goal does not necessarily equate to broader stakeholder utility unless stakeholder concerns are formally embedded in the system's design. This form of goal-orientation may lead to partial or asymmetrical utility maximization, favoring the delegator's interests while neglecting those of downstream stakeholders (e.g., users, employees, customers, vendors, etc.). Therefore, while goal orientation reflects utilitarian logic within the delegator's scope, the absence of stakeholder-inclusive utility considerations reveals the importance of extending the utilitarian calculus beyond mere performance optimization (Floridi et al., 2018; Jones et al., 2018).

While minor cost-benefit trade-offs may exist in ensuring the ethical implementation of agentic IS (Zuboff, 2023), these are generally insignificant. Organizations adopting agentic IS have no inherent conflict between pursuing goals and ethical considerations (Herm et al., 2023). However, the efficiency of agentic IS is often perceived separately from its ethical implications. This distinction may stem from the cognitive processes discussed in dual-process theory (Kahneman, 2011) and bounded rationality (Simon, 1955). Adopting agentic IS requires minimal cognitive effort due to its goal orientation, while evaluating its ethical considerations demands deeper engagement and long-term thinking. Thus, most organizations often prioritize goal orientation over ethical concerns (Hagendorff, 2020; Mittelstadt et al., 2016). Some view ethical considerations as obstacles to monetizing AI (Hagendorff, 2020), leading to criticism from stakeholders who feel that their concerns are ignored.

A recent lawsuit against Wells Fargo alleges that the bank delegated the mortgage decision to its automated underwriting system (CORE) with inadequate human oversight. The plaintiff contends that the CORE algorithm and machine learning components exhibit racial bias (Northern District of California, 2025). This case demonstrates that organizations' decisions to delegate to agentic IS are primarily driven by goal orientation and performance improvements, often neglecting stakeholders' ethical interests in the utility analysis.

Most, if not all, of these ethical concerns have their roots in deontological ethics, a field of ethics that assesses the morality of an action based on its inherent rightness or wrongness (John-Mathews et al., 2022; Leicht-Deobald et al., 2019; Martin et al., 2019; Seele & Schultz, 2022). Stakeholders primarily evaluate agentic IS based on deontological ethics. However, agentic IS's algorithmic logic is rooted in utilitarianism or teleological ethics. Competing ethical priorities associated with agentic IS continue to present challenges. Consequently, stakeholders—whether employees, consumers, users, or others—experience a dilemma that pits goal orientation against ethical considerations. We argue that stakeholders appreciate agentic IS's efficiency and performance enhancements while prioritizing ethical agents free from bias, discrimination, privacy concerns, and similar issues. Therefore, we suggest that stakeholders adopt a mixed approach of deontological and teleological ethics when delegating tasks to agentic IS. Furthermore, the utility of agentic IS can encompass ethical dimensions, such as fairness, privacy, and transparency, rather than focusing solely on performance, which is often overlooked in discussions surrounding AI's potential benefits and drawbacks.

Although agentic IS promise significant utility gains through goal orientation, stakeholders increasingly judge these systems based on performance metrics and their adherence to ethical norms. Empirical evidence underscores this shift: over 75 % of users express concerns about AI-generated misinformation, and 70 % worry about AI's applications in business contexts (Forbes Advisor, 2023), reflecting a growing deontological impulse to assess agentic IS by their intrinsic rightness rather than mere utility. Concurrently, more than 80 % of organizations report skepticism toward AI, a barrier that impedes the realization of agentic IS's teleological benefits (Forbes, 2021). Longitudinal data further reveal that user apprehension rose from 37 % in 2021 to 52 % in 2023 (ComputerWorld, 2023), suggesting that users' bounded rationality heuristics and affective responses increasingly foreground ethical concerns over efficiency alone. Despite this, research on the preferences of users or stakeholders in agentic IS remains nascent (Herm et al., 2023; Kushwaha et al., 2023; Schmitt et al., 2023; Shin et al., 2022). Disregarding these ethical considerations not only jeopardizes reputation and resource allocation but also leads to project failure, as evidenced by notable AI system retractions in recruitment and welfare enforcement (Crockett et al., 2021; Pethig & Kroenung, 2022; Rinta-Kahila et al., 2021; Shellenbarger, 2019).

This research addresses three critical questions that must be considered by organizations implementing agentic IS. How do stakeholders ethically evaluate different levels of agentic IS autonomy in the delegation process? Second, to what degree do stakeholders prioritize goal orientation over ethical considerations when delegating to agentic IS? Third, how does integrating ethics-driven endowments influence the ethical evaluations of agentic IS delegation by stakeholders? By addressing these questions, we provide organizations with actionable insights for responsible and ethical AI implementation that respects stakeholder preferences.

Our results demonstrate that agentic IS stakeholders exhibit both goal orientation and ethical consideration. Furthermore, our research demonstrates that when these systems exhibit attributes that emphasize ethical considerations, stakeholders are more likely to perceive agentic IS and the process of delegating tasks to them as ethical. This underscores the significance of communicating the ethical attributes of agentic IS to stakeholders proficiently. Furthermore, our research findings indicate that stakeholders prefer agentic IS that function under human oversight rather than entirely autonomous systems.

The paper proceeds as follows: First, we establish the theoretical foundation by examining normative ethics theories, delegation concepts, and how organizations often prioritize performance over ethics in agentic IS implementations. Next, we present our research hypotheses by developing a conceptual model that integrates goal orientation with ethical considerations. We then describe two complementary experiments—the first examines how ethics-driven endowments affect ethical evaluations, and the second focuses on how different levels of delegation influence stakeholders' ethical judgments. After presenting our findings, we discuss the theoretical and practical implications for organizations seeking to implement AI responsibly while maximizing stakeholder acceptance.

2. Background

When delegating tasks to agentic IS, organizations frequently prioritize goal orientation over ethical considerations. The goal orientation adheres to utilitarian or teleological ethics. In contrast, the ethical considerations pertain to deontological ethics, leading to conflicting ethical priorities among the key stakeholders involved in the delegation decision or evaluation of agentic IS delegation. The teleological and deontological perspectives are examined within Hunt and Vitell's theory (1986).

This section presents the theoretical foundation of our study. We begin by examining different theories of normative ethics that inform ethical judgments of agentic IS and the delegation to agentic IS. Next, we explore the concept of delegation and its associated ethical responsibilities in the context of agentic IS. Then, we discuss how organizations often prioritize performance over ethics in delegating tasks to agentic IS, creating a potential disconnect with stakeholder preferences. Finally, we present the mixed deontological-teleological ethical framework that informs our research model.

2.1. Different normative ethics theories

A moral standard or norm applies to all moral agents if it can be used to evaluate the character and behavior of all moral actors. The development and justification of such a system is known as *normative ethics*, while the system itself is the *normative ethical system* (Taylor, 1975). The goal of normative ethics is to answer the following questions: What are our fundamental moral obligations? Does the end justify the means, or are there some activities that should be undertaken under no circumstances? (Shafer-Landau, 2012)

Divergent opinions exist regarding the precise ethical force of an action, rule, or disposition. Virtue ethics, deontological ethics, and consequentialism (teleological ethics) are three conflicting viewpoints on how moral questions should be solved, along with hybrid positions that combine these components. Virtue ethics concentrates on the personalities of the actors. Deontological and teleological ethics, in contrast, focus on the status of the action, rule, or disposition (Nodelman et al., 1995).

Deontology is derived from the Greek terms for duty (deon) and science (logos). Deontology is a normative ethical framework that assesses an action's morality based on its inherent rightness or wrongness, guided by a set of rules and obligations, independent of the consequences that may arise (Dierksmeier, 2013).

Teleology is derived from the Greek word telos, which means "end" or "purpose." In teleological ethics, an action is neither good nor bad in and of itself; its consequence makes it either good or bad. Teleological ethics or consequentialism prioritizes the outcome's significance. The moral value of an action is determined by its consequences, with the most favorable action being the one that generates the highest overall value. This prompts an inquiry into the beneficiaries of outcome maximization. Various ethical perspectives emphasize maximizing beneficial outcomes for oneself, others, individuals, and society. The principle of the greatest good for the greatest number encapsulates utilitarianism, a consequentialist ethical theory that assesses right and wrong based on outcomes. Utilitarianism posits that the most ethical decision maximizes happiness for the greatest number of individuals (Graham, 2004).

Understanding these ethical frameworks is crucial because they inform how stakeholders evaluate agentic IS and the decisions made by these systems. While agentic IS are typically designed with utilitarian or teleological principles that maximize outcomes, stakeholders often employ deontological considerations when evaluating the delegation to these systems, creating potential tension between design objectives and stakeholder expectations.

2.1.1. Delegation and ethical responsibility

In management literature, delegation is generally discussed in the context of participatory decision-making (Strauss, 1963; Yukl, 1981) and measured as a continuum of subordinate participation with autocratic decision-making on one end and subordinate decision-making on the other (Bass & Valenzi, 1974; Heller & Yukl, 1969; Vroom & Yetton, 1973). Heller and Yukl (1969) claimed that "delegation refers to decisions that the manager allows subordinates to make on their own" (Heller & Yukl, 1969, p. 230). In contrast, Strauss noted that delegation "is to be contrasted with situations where superiors make decisions either alone or jointly" (Strauss, 1963, p. 70).

In the IS literature, however, the "use" of technology has been the primary study stream for decades, focusing on how individuals utilize IS to achieve their objectives (Burton-Jones et al., 2020). Although the term "delegation" was not explicitly used, early work on computer-based decision support or expert systems intended to assist managers in business-oriented decision making did not exclude the possibility of some levels of delegation (e.g., Bonczek et al. 1979, 1980a, 1980b).

Baird and Maruping (2021) first posited the concept of delegating tasks to agentic IS. Their seminal work focused on furnishing a delegation mechanism for agentic IS that provided key concepts such as agentic IS endowments and human agent endowments. We extend the delegation mechanism of Baird and Maruping (2021) to include the ethical ramifications of agentic IS delegation. Delegating intricate decision-making to agentic IS introduces considerable ethical and legal dilemmas that require thorough examination (Candrian & Scherer, 2022). Delegating tasks to agentic IS does not absolve the delegator of their ethical responsibilities. Consequently, we contend that delegation constitutes the essential mechanism through which ethical dilemmas associated with agentic IS arise, and ethical considerations must be integrated into the delegation process.

2.1.2. Prioritization of performance over ethics in agentic IS delegation

Collins et al. (2021) categorized AI to support three business needs: process automation, cognitive insights, and cognitive engagement, emphasizing that AI is goal-oriented. Organizations implementing agentic IS report significant revenue growth and cost reductions (Fosso Wamba, 2022; Stanford AI Index, 2023). Some examples of agentic IS that provide utility and efficiency include algorithmic automation in HRM practices (Cheng & Hackett, 2021), the use of chatbots to improve performance in the service industry (Hill et al., 2015), and the use of machine learning to predict corporate fraud (Xu et al., 2022).

Although AI utility can include ethical dimensions (e.g., fairness, privacy, and transparency), most organizations prioritize goal orientation over ethical considerations for agentic IS implementations. Bostrom and Yudkowsky (2018) proposed that this phenomenon may stem from competitive advantage, in which firms prioritize innovation over ethical considerations. The slow progression of AI legislation has resulted in a lack of mechanisms to compel organizations to adhere to ethical principles (Hagendorff, 2020). Without enforcement mechanisms, organizations face no significant threat of prioritizing ethics aside from potential reputational damage (Hagendorff, 2020). Consequently, many organizations are attempting to leverage AI for financial gain in various applications, primarily driven by economic incentives rather than ethical considerations. Moreover, AI engineers and developers frequently lack the education or authority to tackle ethical issues. Recent examples of performance prioritization without carefully examining ethical implications include facial recognition systems deployed without adequate bias testing (Buolamwini & Gebru, 2018) and medical diagnostic systems trained on non-representative patient populations (Obermeyer et al., 2019). These examples highlight the ethical risks when performance is prioritized over ethical considerations. Thus, AI applications frequently deviate from ethical standards and principles, resulting in insufficient ethical considerations throughout the process (Hagendorff, 2020; Mittelstadt et al., 2016).

2.2. Ethical dilemma of agentic IS

Literature on the user or stakeholder perception of AI is scarce (Herm et al., 2023; Wanner et al., 2022), which leads to theoretical assumptions regarding the compromises between efficiency and ethicality (Arrieta et al., 2020). Herm et al. (2023) empirically disproved one such assumption about performance and explainability. Shin (2021) found that trust in AI plays a key role in perceived performance, highlighting the importance of human-centered AI. Shin and Park (2019) highlight the importance of understanding the cognitive processes of users and their ability to reflect these processes in algorithm designs.

We posit that stakeholders appreciate AI's efficiency and performance gains, but not at the expense of ethics. The predominant motivation for stakeholders to embrace AI lies in pursuing enhanced efficiency; however, when the quest for efficiency intersects with ethical considerations, they delineate clear boundaries (Chalhoub & Flechais, 2020; Ryan, 2020). Therefore, we argue that when delegating tasks to AI, stakeholders utilizing agentic IS are concerned with goal orientation and ethical implications, adhering to a mixed deontological-teleological framework.

2.3. Mixed deontological-teleological ethical system

Hunt and Vitell (1986) proposed a general theory of marketing ethics that attempts to explain the decision-making process in ethically problematic situations. Hunt and Vitell (1986) suggest that creating a comprehensive system of deontological rules that contains a finite number of exceptions, without conflicts, appears impossible (Hunt & Vitell, 1986, p. 7).

Hunt and Vitell (1986) argued that utilitarianism faces three significant challenges for its proponents. First, determining which benefits should be maximized is a significant challenge. Maximizing the greatest good for the greatest number is also a significant challenge. Additionally, measuring the amount of good realized across various outcomes and people with different utility functions is challenging. Lastly, many ethicists believe that maximizing the total good produced may not always lead to a morally correct solution, as the total good may be distributed unjustly (Hunt & Vitell, 1986, p. 7). Such issues have led many moral philosophers, including Frankena (1963), to recommend a mixed deontological-teleological system of ethics. Hunt and Vitell (1986) argued that their model, which is now treated as a theory, accounts for both the deontological and teleological aspects of ethics. Moreover, unlike the previous models, the H-V model is empirically testable.

The central hypothesis of the H-V model posits that both deontological and teleological evaluations inform consumers' ethical judgment, which subsequently influences their intentions and behaviors. Additionally, deontological and teleological assessments consider perceived alternatives, whereas only teleological assessments consider perceived consequences. In deontological evaluation, the individual assesses the intrinsic rightness or wrongness of the behaviors associated with each alternative. The process entails comparing behaviors against a defined set of deontological norms that reflect personal values or behavioral guidelines. The model explains that the desirability and likelihood of outcomes, along with the significance of stakeholders, affect teleological evaluation. The outcome of the teleological assessment involves beliefs regarding the relative goodness or badness associated with each alternative as perceived by the individual.

The H-V model has been extensively tested in various studies. Vitell and Hunt (1990) found that managers rely on deontological and teleological factors when making ethical judgments. Mayo and Marks (1990) experimented with marketing researchers and found that deontological and teleological evaluations jointly determine ethical judgments to resolve dilemmas. Hunt and Vasquez-Parraga (1993) used a field experimental design to examine the responses of 747 sales and marketing managers to ethical issues involving salespeople. In the situations studied, deontological considerations, rather than teleological considerations, heavily influenced marketers' ethical judgments and their plans for acting. In a consumer context, Vitell et al. (2001) discovered that while both deontology and teleology were important in decision making, consumers tended to rely more on deontology (ethical norms) than on teleology (consequences) when making ethical judgments and intentions.

We posit that stakeholders utilizing agentic IS are cognizant of both goal orientation—aligned with the teleological —and ethical considerations—rooted in the deontological perspective of ethics. Hunt and Vitell (1986) argued that these evaluations are not mutually exclusive; instead, individuals evaluate both deontological and teleological perspectives. For instance, a healthcare provider might justify delegating to AI for preliminary screenings (teleological) if it adheres to transparency protocols (deontological), illustrating how principles and outcomes jointly shape ethical intent. Similarly, an HR manager might delegate resume screening to AI (teleological) if the AI system adheres to privacy, justice, and fairness.

The relative emphasis on deontological versus teleological factors is dependent on contextual factors. In low-risk circumstances, we propose that individuals adhere to a mixed deontological-teleological perspective when delegating to agentic IS. Individuals tend to lean more toward deontological ethics that emphasize principle-based evaluation in situations characterized by significant risks to human well-being. In contrast, individuals tend to be more teleological in extreme ethical dilemmas, such as the trolley problem that involves choosing to divert a runaway trolley to kill one person instead of five (Thomson, 1985). Thus, we propose that the overall ethical judgment of delegation to agentic IS is a function of delegation's deontological and teleological evaluation.

Much research has been conducted to underscore the benefits or positive aspects of AI (see e.g., Miller, 2018; Pezzo & Beckstead, 2020; Silverman & Waller, 2015). Similarly, a large corpus of research, frequently referred to as the "dark side of AI," underscores the ethical concerns associated with AI (see e.g., Akter et al., 2021; Luengo-Oroz et al., 2021; Martin, 2022). Appendix A in the supplementary materials provides a detailed literature review of ethical issues related to agentic IS. Nevertheless, to the best of our knowledge, the combination of the utility or bright side with the ethical considerations or the dark side has not been investigated. H-V theory is of particular significance because it establishes the essential framework for the mixed deontological-teleological evaluations, which are essential for comprehending how stakeholders reconcile the utility of AI with the ethical concerns of AI delegations.

3. Theoretical foundation and research model

Building on the theoretical foundation discussed above, a comprehensive model of how stakeholders evaluate agentic IS from goal-oriented and ethical perspectives is developed. Our model integrates three key elements: (1) the endowments embedded in agentic IS (both performance- and ethics-driven), (2) the level of delegation or autonomy granted to agentic IS, and (3) the ethical evaluation process that delegators and stakeholders employ when assessing agentic IS and delegation decisions. This integrated approach enables us to examine how stakeholders balance utility with ethical considerations when interacting with agentic IS.

3.1. Conceptual model

The conceptual model focuses on goal orientation and ethical consideration, as shown in Fig. 1. Our model focuses on the expansive role of human delegators by incorporating key stakeholders in the delegation decision within the context of agentic IS's organizational implementation. These components are discussed next.

3.1.1. Incorporating ethical considerations with goal orientation

Simon (1955), (1956) emphasized the constraints faced by human decision-makers resulting from information overload and uncertainty in decision-making. Cognitive and social psychologists have identified dual processing as a crucial mechanism for managing information overload. This involves both detailed, deliberative analysis and the capacity to process information efficiently with reduced cognitive effort (e.g., Evans, 1984; Schneider & Shiffrin, 1977). This led to the development of dual-process theories concerning reasoning, social cognition, and decision-making (Chaiken & Trope, 1999; Evans, 2003, 2006; Schneider & Shiffrin, 1977; Smith & Levin, 1996). In management literature, dual process theory has been employed to elucidate organizational decision-making processes. Kahneman and Frederick (2012) proposed that human information processing occurs through two distinct yet complementary mechanisms: one that operates largely outside of conscious awareness and control (automatic processing) and the other that functions within the scope of conscious awareness and control (controlled processing). Automatic processing allows individuals to swiftly and easily navigate extensive information, whereas intentional controlled processing requires thorough analysis.

Humans often view the efficiency or goal orientation of agentic IS in a manner distinct from its ethical implications, a divergence influenced by cognitive mechanisms as outlined in dual process theory (Kahneman, 2011; Kahneman & Frederick, 2012) and bounded rationality (Simon, 1955). Dual process theory delineates two distinct modes of cognition: Type 1, characterized by its rapid, intuitive, and automatic nature, and

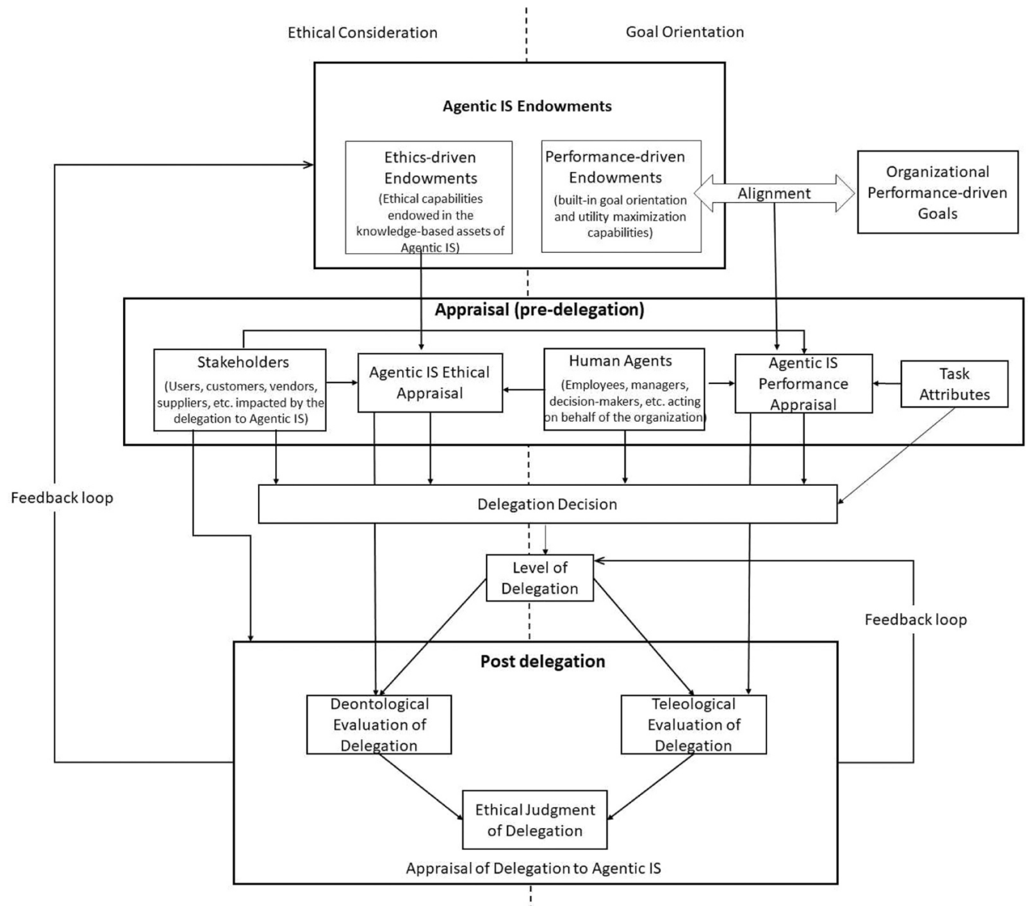

Fig. 1. Conceptual model of ethical appraisal of agentic information systems and delegation.

Type 2, which is characterized by a slower, more deliberative, and analytical approach (Kahneman, 2011). Efficiency metrics, including the AI system's speed or accuracy, correspond with intuitive reasoning as they provide prompt, discernible results that necessitate little cognitive exertion for assessment. For example, users instinctively regard an AI chatbot as efficient when it addresses inquiries with remarkable speed, often without engaging in further scrutiny. Conversely, ethical considerations—such as algorithmic bias, privacy violations, or accountability gaps—necessitate the involvement of deeper cognitive processes, as they encompass abstract values, long-term implications, and intricate trade-offs. The complexities of these issues demand ongoing intellectual engagement and specialized knowledge, resulting in undervaluation of ethical considerations in favor of efficiency (Brundage et al., 2020; Kahneman, 2011).

The concept of bounded rationality further exacerbates the disparity in understanding, which suggests that the availability of information limits human decision-making, the constraints of time, and the capacity of cognitive processes (Simon, 1955). Efficiency improvements, exemplified by claims such as "AI reduces costs by x percentage," present clear and compelling evidence that aligns seamlessly with heuristic-driven evaluations. Ethical risks, in contrast, frequently present themselves as abstract or probabilistic harms (e.g., "AI may perpetuate systemic biases"), which can be more challenging to prioritize within the cognitive capacity limits (Simon, 1955). Although the rapidity of a hiring algorithm may be lauded, its capacity to perpetuate discrimination rooted in historical data necessitates a thorough examination that often escapes the attention of users or developers until a significant issue emerges. This imbalance illustrates a propensity to favor concrete, immediate advantages rather than abstract, postponed dangers—a defining characteristic of limited rationality (Cave & Dihal, 2019; Simon, 1955).

To address this gap, our model utilizes a value-sensitive design approach (Friedman et al., 2013) that renders ethical considerations as prominent as AI efficiency metrics, thereby prioritizing ethics-driven characteristics in developing agentic IS. Additionally, our model enables stakeholders to assess agentic IS's performance and ethical dimensions throughout and following the delegation process. This encourages organizational decision-makers to engage in thorough, structured, and systematic thinking, stimulating Type 2 processes (Hodgkinson & Sadler-Smith, 2018), before choosing a specific course of action regarding AI.

3.1.2. *The expansive role of humans in agentic IS delegation*

Baird and Maruping (2021) characterized delegation as a dyadic relationship between the human and agentic IS. The ethical ramifications of such delegation involve stakeholders directly or indirectly affected by the delegation to agentic IS. Like many other lawsuits involving AI, the plaintiff in the Wells Fargo case is a user affected by the organization's decision to delegate a task to AI. Thus, the concept of humans in agentic IS delegation is expansive, involving key stakeholders, including consumers, clients, employees, users, and others directly or indirectly impacted by the delegation to agentic IS (Zuboff, 2023). Fig. 1 highlights this expansive role; the human agent, as shown in Fig. 1, is the organizational delegator, such as an employee, manager, or decision-maker, acting on behalf of the organization implementing agentic IS. In contrast, stakeholders (such as customers, vendors, users, and suppliers) represent the parties impacted by the delegation to agentic IS.

This expansive role of humans in agentic IS delegation aligns with the stakeholder theory (Freeman, 1984). According to the theory, stakeholders are any individual or group with a vested interest in or impacted by the corporation (Carroll, 1989; Freeman, 1984). Frequently, tension arises between agency and stakeholder theories, rooted in a clash between economic perspectives that overlook the moral dimensions of contemporary corporations and ethicists who underscore the necessity of comprehending ethical ramifications within business ethics (Bowie & Freeman, 1992). Therefore, the stakeholder theory is in harmony with ethical considerations regarding the conflict between goal orientation and ethical considerations.

While the model developed incorporates the delegator's and stakeholders' perspectives, in situations where individuals may occupy dual roles (e.g., a manager who delegates recruitment to AI but is also subject to AI-based performance evaluation), the salience of these mechanisms shifts depending on their role, with potential tensions arising when efficiency-oriented delegator's preferences conflict with fairness-oriented stakeholder's concerns.

3.1.3. Agentic IS endowments

Baird and Maruping defined endowments as intrinsic “resource-based assets (e.g., knowledge) and capabilities (e.g., ability to think abstractly)” (Baird & Maruping, 2021, p. 324). Although the term “endowment” may be new, the concept is not. Hess et al. (2000) proposed that an agent should possess three essential qualities: goal orientation, persistence in achieving goals, and reactivity within the environment it operates. Building on the work of Baird and Maruping (2021), we introduce the concept of endowments, categorizing these intrinsic assets and capabilities into two distinct types: performance-driven endowments and ethics-driven endowments.

Agentic entities possess knowledge-based assets that allow them to evaluate their surroundings and pursue their objectives. These knowledge-based assets contain algorithms that can analyze a dataset to infer and produce results. An algorithm is a sequence of computational operations that transforms inputs into outputs (Cormen, 2009) and enhances both efficiency and accuracy (Seaver, 2017). In our model, these knowledge-based assets are referred to as *performance-driven endowments*.

Knowledge-based assets may also include ethical capabilities endowed during the agentic IS lifecycle's build or training phases (Martin, 2019). For example, the training dataset can be treated for historical biases, fairness, and prejudice, and the data can be gathered with human consent while human dignity and privacy are protected. Different algorithmic techniques can be endowed in agentic IS to address ethical concerns. For instance, agentic IS can be equipped with de-identification algorithms to conceal personally identifying information. Similarly, k-anonymity, a technique that anonymizes each record relative to at least k-1 other records, can be used for anonymization. Agentic IS can de-bias the data (Zemel et al., 2013) by over-sampling from minority classes (Chawla et al., 2002) or using adaptive synthetic sampling (He et al., 2008) to address fairness, bias, and discrimination-related issues. Adhering to these ethical principles improves the reliability, safety, and trustworthiness of agentic IS, resulting in improved adoption (Shneiderman, 2020). In our model, such endowments are referred to as *ethics-driven endowments*.

From a developmental standpoint, direct tensions exist between ethics- and performance-driven endowments in agentic systems. Implementing fairness-aware algorithms to reduce prejudice against specific groups often involves modifying model predictions to achieve equitable outcomes, potentially compromising overall predictive accuracy. Fairness constraints, such as demographic parity and equal opportunity, can reduce a model's accuracy by favoring equitable treatment over optimal prediction (Corbett-Davies et al., 2023). Similarly, privacy-preserving methodologies, such as differential privacy, unavoidably limit the volume or specificity of data accessible for training, thus diminishing model efficacy and performance. For instance, introducing noise to guarantee differential privacy can conceal detailed patterns in the data, resulting in diminished predictive accuracy (Abadi et al., 2016; Dwork & Roth, 2013). These examples illustrate that while ethics-driven interventions enhance ethical results, they may compromise performance, necessitating a careful assessment of trade-offs in system design. The tension between performance and ethics is ongoing for users and stakeholders, as well as being inherent in the design and architecture of AI systems.

3.1.4. Incorporating mixed deontological-teleological ethics into agentic IS delegation

Performance-driven endowments are designed to pursue objectives and optimize utility. Agentic IS are designed to conform to utilitarian principles, emphasizing the maximization of benefits for the majority (Russell & Norvig, 2021). The model illustrates this goal orientation and utilitarianism (or teleology) on the right side, as shown in Fig. 1.

Similar to other information systems, integrating agentic IS within an organization begins with a thorough appraisal process. This process involves examining and comparing software solutions to ensure that they align with technical, operational, and business requirements. Organizational performance objectives are aligned with agentic IS capabilities to evaluate fit for designated tasks. In our model, this is referred to as *agentic IS performance appraisal*. This aspect of the model highlights the utilitarian assessment for the agentic IS and the delegation to the agentic IS.

Our model highlights the significance of ethics-driven endowments in the development of agentic IS. It encourages Type 2 processing, as outlined by Hodgkinson & Sadler-Smith (2018), to assess both ethics- and performance-driven endowments. The left side of the model incorporates ethics-driven endowments and involves key stakeholders in assessing how these endowments are manifested. In alignment with how stakeholders assess agentic IS for ethical considerations, this side of the model emphasizes the deontological assessment of the agentic IS and the deontological evaluation of delegation.

Delegation refers to the process of transferring rights and responsibilities for the execution of tasks and their outcomes to another party (Baird & Maruping, 2021, p. 317). Delegation to agentic IS can occur at various levels, as illustrated by the *level of delegation* within the model. The pre-delegation phase, which encompasses performance and ethical appraisal, is succeeded by the delegation decision, which is subsequently succeeded by the delegation of rights and responsibilities. Consequently, the model assesses the ethical judgment of the delegation using a mixed deontological-teleological ethical framework, as proposed by the H-V theory (1986).

Lastly, the appraisal of agentic IS delegation, which involves the delegation's performance and ethical evaluation, informs the agentic IS endowments and the delegation level for subsequent uses of the agentic IS. These are represented as feedback loops in the model, as shown in Fig. 1.

3.2. Construct definitions

The definitions for the constructs used in this study are derived from the extant literature and are presented in Table 1.

3.3. Research model and hypotheses development

The research model used in this study is grounded in the agentic IS delegation framework (Baird & Maruping, 2021), H-V theory of ethics (Hunt & Vitell, 1986), and concepts from agency theory (Jensen & Meckling, 1976) and stakeholder theory (Freeman, 1984). The research model, as shown in Fig. 2, also uses the dual process theory (Kahneman, 2011; Kahneman & Frederick, 2012) to represent goal orientation and ethical consideration as the two underlying processes in evaluating performance- and ethics-driven endowments during the delegation process. The hypotheses marked in Fig. 2 indicate the relationships tested in this study; the links that do not indicate a hypothesis are controlled in this study. While our conceptual model provides a

Table 1
Definitions of constructs.

Construct	Definition
Agentic IS endowments	Intrinsic knowledge-based assets that are integrated in the construction of agentic IS.
Performance-driven endowments	Performance-driven capabilities endowed by knowledge-based assets in the agentic IS (e.g., what is the algorithmic logic, what data sources or variables are used, what are the factors and their corresponding weights, and what performance gains does the model provide).
Ethics-driven endowments	Ethics-driven capabilities endowed by knowledge-based assets in the agentic IS (e.g., how the data was acquired to build the model, how the model was trained, and how the training dataset was treated for bias and fairness).
Agentic IS performance appraisal	The evaluation of the agentic IS's performance-driven endowments to assess their performance fit by the human agent (delegator) or stakeholders.
Agentic IS ethical appraisal	The evaluation of the agentic IS's ethics-driven endowments to assess their ethicality by the human agent (delegator) or stakeholders.
Level of delegation	Perceived extent to which rights are transferred between human and agentic IS (e.g., manual control, human selection of decision, autonomous control, etc.).
Deontological evaluation of delegation	The ethical assessment of delegating a task or a set of tasks to an agentic IS, focusing on intrinsic morality and adherence to ethical principles within a specific scenario.
Teleological evaluation of delegation	The ethical assessment of delegating a task or a set of tasks to an agentic IS, focusing on utility maximization and goal orientation within a specific scenario.
Ethical judgment of delegation	Ethical judgment of delegation is a function of deontological and teleological evaluations of delegation within a specific scenario.

The research model and its associated hypotheses are discussed in the next section.

comprehensive framework for understanding ethical and performance evaluations of agentic IS, our empirical research focuses on testing a subset of relationships within the conceptual model. Specifically, our research model examines how ethics-driven endowments and different levels of delegation influence stakeholders' ethical judgments through deontological and teleological evaluation processes. Next, we discuss the hypotheses.

Agentic IS can incorporate ethics-driven endowments by establishing features that safeguard privacy and confidentiality during data collection, and ensuring the data is unbiased, equitable, and discrimination-free. By addressing these issues, ethics-driven endowments enhance the ethical evaluation of agentic IS by human stakeholders. Shneiderman (2020) posits that ethically designed agentic IS are perceived as safer, more reliable, and trustworthy, leading to improved adoption rates. These perceptions will significantly enhance human performance and foster self-efficacy, mastery, creativity, and responsibility (Shneiderman, 2020).

Signaling theory, first articulated by Spence in 1978, offers a comprehensive framework for understanding how individuals and organizations communicate information to mitigate information asymmetries in decision-making contexts. Grounded in economic principles, the theory outlines that agents use observable indicators—such as credentials, certifications, or strategic actions—to effectively convey unobservable attributes to decision-makers (Spence, 1978). For example, candidates may demonstrate their capabilities in labor markets through educational credentials, allowing employers to navigate hiring choices amid uncertainty. This principle transcends individual interactions within the job market, permeating organizational contexts where firms convey quality to shape stakeholder decisions through branding, pricing strategies, or sustainability initiatives (Connelly et al., 2011; Plummer et al., 2016). Signaling theory has a profound influence in numerous domains of organizational studies, including leadership, marketing, strategic management, and human resources (see e.g., Allison et al., 2013; Appels, 2023; Erdem & Swait, 1998; Guest et al., 2021). The

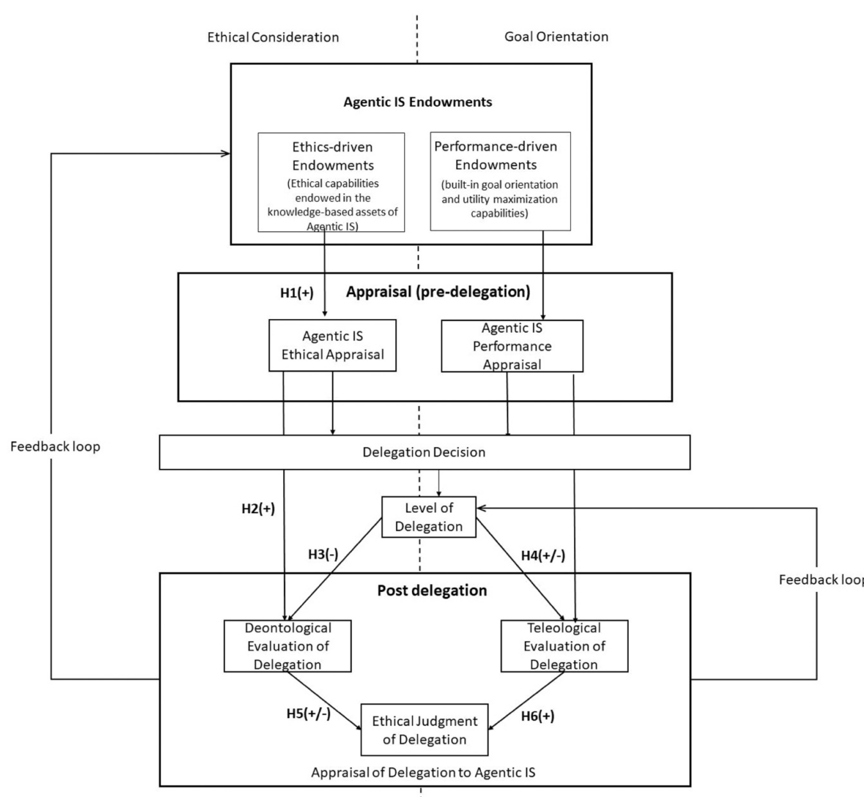

Fig. 2. Technology ethical assessment model (TEAM) with hypotheses.

importance of signaling in terms of AI has recently been discussed in the AI-human team interactions by Smith et al. (2025).

By incorporating ethics-driven endowments that treat the agentic IS for bias, discrimination, privacy, fairness, and other ethical issues, and by signaling the presence of these ethics-driven endowments, stakeholders are more likely to have a higher ethical appraisal of agentic IS. As a result, we put forward the following hypothesis:

H1. : Ethics-driven endowments in agentic IS positively influence agentic IS ethical appraisal

A positive ethical appraisal of the agentic IS inspires assurance and trust among human stakeholders in the ethical capabilities and endowments of the agentic IS (Shneiderman, 2020). Consequently, the deontological assessment of delegation to the agentic IS is enhanced. According to agency theory, if the principal has confidence in the capabilities of an agent, the principal is more inclined to delegate authority to the agent (Casadesus-Masanell, 2004; Cuevas-Rodríguez et al., 2012). Therefore, we contend that the ethical evaluation of agentic IS positively impacts the deontological evaluation of delegation to agentic IS. Trust is the foundational mechanism, which is fostered by ethics-driven endowments between the human stakeholders and agentic IS (Shneiderman, 2020).

The ethical standards of software thoroughly document the connection between trust and the inclination to delegate tasks to it. Studies have demonstrated that confidence in a system's ethical congruence—anchored in the perceptions of fairness, transparency, and adherence to moral standards—profoundly affects delegation practices (Hancock et al., 2011). When stakeholders regard software as ethically sound, they are more inclined to consider it reliable and competent, which are essential precursors to trust in automation (Lee & See, 2004). This confidence encourages delegation of responsibilities, as individuals tend to favor systems they trust to operate in alignment with collective principles (Shin, 2021). Furthermore, when there is a high level of ethical trust, evaluations conducted after delegation tend to exhibit a favorable bias, as users mentally associate successful outcomes with the moral integrity of the system (Veale et al. 2018). Empirical investigations into algorithmic decision-making reveal that ethical transparency alleviates skepticism, thereby improving both delegation levels and user satisfaction (Langer et al., 2022). Consequently, ethical trust is an essential intermediary in the delegation process, influencing both initial dependence and subsequent assessments of technological systems. Thus, we hypothesize the following:

H2. : A more positive agentic IS ethical appraisal is associated with increased deontological evaluation of delegation

Delegation to agentic IS has different levels; these levels are represented as archetypes ranging from reflexive (reactive) to prescriptive (autonomous) by Baird & Maruping (2021). The applied ergonomics literature proposes similar levels of delegation (see Vagia et al. 2016). These levels of delegation represent increasing levels of autonomy, which in turn diminish human oversight in the delegation mechanism.

The overall consensus in the extant literature on AI is that human oversight should be maintained in algorithmic decision making (Martin, 2019; Martin & Waldman, 2022). This also aligns with the European Commission's ethical guidelines for trustworthy AI (European Union, 2019). Empirical studies indicate that when human experts maintain final decision-making authority, individuals exhibit a greater propensity to trust and accept AI recommendations, a phenomenon referred to as the "human-in-the-loop" (HITL) bias (Binns et al., 2018; Dietvorst et al., 2015; Shneiderman, 2020). This preference arises from the belief that human oversight reduces the risks of algorithmic errors or biases. Dietvorst et al. (2015) demonstrated that users resist fully autonomous systems, preferring hybrid models in which humans validate or modify algorithmic outputs. Based on the extant literature on AI, we propose that the higher the delegation level (more autonomous, involving lesser human involvement), the lower the deontological evaluation of delegation to agentic IS by stakeholders. Therefore, we hypothesize the following:

H3. : A more positive level of delegation to agentic IS (involving lesser human involvement) is associated with a decreased deontological evaluation of delegation.

The utility of agentic IS is always centered on goal orientation. Organizations use agentic IS for process automation, cognitive insights, and cognitive engagement (Collins et al., 2021). Organizations implementing AI report significant revenue growth and cost reductions (Fosso Wamba, 2022; Stanford AI Index, 2023). The use of AI in HRM practices (Cheng & Hackett, 2021), chatbots (Hill et al., 2015), and task automations (Xu et al., 2022) suggests that organizations seek to maximize efficiency by using AI at the highest levels of automation. Thus, higher automation levels should positively impact the teleological evaluation of delegation.

However, in an ethical situation, teleological evaluation is a function of perceived consequences, the probability and desirability of the consequences, and the importance of these consequences for the stakeholders (Hunt & Vitell, 1986). Varying levels of delegation for agentic IS impact how human stakeholders perceive delegation outcomes. These delegation outcomes convey the implications of delegation to human stakeholders, allowing them to identify the most significant benefit for the greatest number of individuals. Empirical studies examining the H-V model indicate that individuals prioritize deontological norms when ambiguous or non-extreme consequences are present before conducting teleological assessments (Hunt & Vasquez-Parraga, 1993). Similarly, Singhapakdi et al. (1996) found that in marketing ethics scenarios, ambiguous consequences prompted decision-makers to prioritize deontological principles, such as fairness, over utilitarian outcomes. O'Fallon and Butterfield (2013) assert that deontological reasoning predominantly influences ethical judgments in low-consequence clarity scenarios.

We propose that the perceived delegation outcomes of human stakeholders influence the delegation's teleological evaluation. However, the relationship could be positive or negative depending on the situation and the severity of perceived consequences. Increased delegation levels, characterized by reduced human involvement in tasks, typically lead to enhanced utility and a more favorable teleological assessment of delegation to agentic IS. However, as an example of severe consequences, if increased levels of delegation lead to a lawsuit (a negative outcome observed in the Wells Fargo case). In this case, it will adversely affect the assessment of delegation to agentic IS. Therefore, we hypothesize the following:

H4. : Level of delegation impacts the teleological evaluation of delegation to agentic IS

Following the logic of goal orientation and ethical consideration and in line with the H-V theory (1986), we posit that a human's ethical judgment of delegation is a function of the individual's deontological and teleological evaluations of delegation. Hunt and Vitell (1986) argued that it is improbable for many individuals to consistently adhere to deontological or teleological ethical frameworks in various situations. The combined deontological-teleological ethical framework has been previously used to assess the ethical or unethical use of IS (Banerjee et al., 1998). The ethical perspective prevailing in the literature on agentic IS is predominantly deontological. However, contemporary agentic IS follow utilitarianism, prioritizing maximizing utility through acts and consequences (Russell & Norvig, 2021). Since our model combines goal orientation with ethical consideration, we posit that agentic IS stakeholders follow a mixed deontological-teleological ethical framework to assess delegation to agentic IS.

As for the sign of the relationships, we hypothesize that the deontological evaluation of ethics will positively influence the ethical judgment in ethical scenarios with mild to moderate consequences. In other words, higher values of deontological evaluation of delegation will result in greater ethical judgment of delegation. In such circumstances, the teleological evaluation will follow the sign of the relationship of the deontological evaluation. In contrast, when the consequences of the outcome are grievous, the teleological evaluation of delegation will predominate, whereas the deontological evaluation of delegation will negatively impact the ethical judgment. The trolley dilemma, which involves diverting a runaway trolley to kill one person instead of five, is often used in academic literature as an extreme moral situation (Thompson, 1976; Thomson, 1985). Medical triage for limited resources during a pandemic exemplifies how adherence to deontological principles, without consideration of survivability, can lead to significantly increased mortality rates (Emanuel et al., 2020). Consequently, we postulate the following:

H5. : Deontological evaluation of delegation influences the ethical judgment of delegation to agentic IS

H6. : A more positive teleological evaluation of delegation to agentic IS is associated with increased ethical judgment of delegation

4. Research method

This section describes the research method used to test the hypotheses presented in Section 3. This research primarily tests the stakeholders' perspective on agentic IS delegation.

4.1. Research design and rationale

Owing to concerns regarding social desirability bias, we examined ethical behavior using scenario-based studies. By placing participants in hypothetical, third-person scenarios or vignettes, these methods establish psychological distance, diminishing the pressure to adhere to normative expectations (Fisher, 1993). Alexander and Becker (1978) assert that vignette-based surveys enable respondents to express their authentic beliefs through fictional characters or scenarios, thereby avoiding self-presentation issues. Scenario-based methods have been effectively used in previous studies dealing with IT ethics (Banerjee et al., 1998).

Two complementary experiments were conducted. Experiment 1 used a between-subjects design to examine how ethics-driven endowments and the level of delegation influence ethical judgments. Experiment 2 used a within-subjects design that focused specifically on the effect of delegation level on ethical judgment, allowing participants to compare different levels of delegation. To evaluate the stakeholder perspective on agentic IS delegation, both experiments asked participants to role-play as job applicants.

4.2. Measurement

To ensure construct validity, all measures were derived from previously validated scales whenever possible. The level of delegation was measured using a five-point scale adapted from the taxonomy devised by Vagia et al. (2016) from the applied ergonomics literature. The taxonomy proposes eight control levels ranging from manual to autonomous. We consolidated and streamlined the levels by aligning them with the four core automation functions identified by Parasuraman et al. (2000): information acquisition, information analysis, decision selection, and action implementation. Additionally, we acknowledged that significant distinctions arise at transitions in decision authority and implementation control. Consequently, we integrated the "computer selects and executes with human approval" and "computer executes and informs" levels into a unified category of computer decision selection, as both uphold system-led actions followed by human notification and retain the same accountability frameworks (Lee & See, 2004). We also removed the conditional notification variants ("informs only if asked" and "informs only if it decides to"), as they do not change the locus of control or improve the situational awareness of users—key components of Endsley's (1995) model—and merely add unnecessary complexity without enhancing construct validity. The five-point scale spans from manual control to fully autonomous operation and delineates key authority and awareness transition points. This framework provides a concise and theoretically sound taxonomy that aligns with empirical findings regarding trust, workload, and accountability in intelligent systems (Parasuraman et al., 2008). Thus, we use the following five levels: a) manual control – agentic IS provides information to human; human has the option to use the information for the decision, b) decision proposal – agentic IS makes recommendations to human; human makes the decision from or outside of the recommendations, c) human selection of decision – agentic IS makes recommendations to human; human decides on the recommendations, d) computer selection of decision – agentic IS makes the decision and informs the human, and e) autonomous control – agentic IS autonomously performs the task and only informs the user if out of specification error conditions occurs.

The agentic IS ethical appraisal items were adapted from Jang et al. (2022)'s attitude toward AI scale. The deontological evaluation of delegation, the teleological evaluation of delegation, and the ethical judgment of delegation are constructs borrowed from H-V theory. Most, if not all, of the theory testing on H-V theory used one item for measuring ethical judgment. For example, Cole et al. (2000) used one item to measure the teleological evaluation of purchasing managers. Similarly, Vitell et al. (2001) used two questions to measure ethical judgment, but used them independently for statistical purposes. However, SEM requires at least three indicators (items) per construct to ensure proper model identification (Kline, 2016; Sarstedt et al., 2021). Therefore, we used four questions each to measure these constructs, two of which were borrowed from Vitell et al. (2001). The constructs are supplemented by two novel items.

A panel comprising two academic and two industry experts examined the questionnaire to assess its face validity after its creation. The scale's content validity was confirmed by conducting a Q-sort exercise with four AI industry experts. Appendix C shows the scale items used in the study. All items were rated using a 5-point Likert scale (Strongly Disagree = 1 to Strongly Agree = 5).

We combined ANOVA and covariance-based structural equation modeling (CB-SEM) to assess our research model comprehensively. ANOVA was selected to test the direct effects of our manipulated variables because it allows for a rigorous examination of group differences in experimental designs (Hair et al., 2018). CB-SEM was chosen over alternative approaches, such as PLS-SEM, because our research focused on testing a theory-driven model with established constructs rather than exploring new relationships or predictions (Rigdon, 2012). Additionally, our sample size was adequate for maximum likelihood estimation. CB-SEM provides more conservative estimates of path relationships than methods such as PLS-SEM while accounting for measurement error, increasing confidence in our findings (Kline, 2016).

We verified that the necessary statistical assumptions were met before conducting our analyses. For ANOVA, we assessed normality using Shapiro-Wilk tests and homogeneity of variance using Levene's test, and we examined Q-Q plots for each dependent variable. Data normality was assessed using the Shapiro-Wilk test. The data did not significantly deviate from a normal distribution ($W = 0.968$, $p = .171$). Therefore, the assumption of normality was met. Additionally, we examined multicollinearity among predictor variables using variance inflation factors (VIFs), with all values below 3, well below the recommended threshold of 10 (Hair et al., 2018).

Missing data were minimal in our study (less than 5 %) and were determined to be completely missing at random (MCAR) based on Little's MCAR test ($\chi^2 = 70.44$, df = 66, $p = .341$). Complete information maximum likelihood (FIML) estimation was employed in our SEM analyses, which has been shown to produce unbiased parameter estimates and standard errors under MCAR conditions (Enders, 2010). Given the low percentage of missing data, we used listwise deletion for the ANOVA analyses.

5. Experiments and results

5.1. Measurement model

The model's reliability is mentioned only once in this section because Experiment 2 uses a subset of the variables used in Experiment 1. The analysis was conducted using the Lavaan utility in RStudio. The items were modeled as reflective indicators of latent first-order factors.

The final model had 15 items: three for agentic IS ethical appraisal, and four for deontological evaluation of delegation, teleological evaluation of delegation, and ethical judgment of delegation. The CFA model demonstrated an excellent model fit. Model χ^2 was 195.87 (df = 103) for a χ^2/df of 1.90, and NFI, CFI, TLI, RMSEA, and SRMR were 0.925, 0.962, 0.95, 0.06, and 0.046, respectively. CFI and TLI values above 0.95, RMSEA below 0.06, and SRMR below 0.08 generally indicate a good fit (Hu & Bentler, 1999). Convergent validity was evaluated for both constructs using the criteria recommended by Fornell and Larcker (1981). The construct reliability ranged from 0.814 to 0.953. The AVE exceeded 0.5 for all the constructs. The discriminant validity was measured using the criteria outlined by Fornell and Larcker (1981), which suggests that the square root of AVE should exceed the correlation between the constructs. The square roots of AVE for all constructs were greater than the correlation between other constructs, except for the deontological evaluation of delegation. This is expected because H-V theory (1986) defines ethical judgment as a function of deontological and teleological evaluations and establishes this as a testable proposition. Thus, for scenarios where most users follow deontological ethics, the ethical judgment correlates highly with deontological evaluation. Prior testing on H-V theory has found similar results regarding the high correlation between deontological ethics and ethical judgment. Cronbach's alpha for all the constructs exceeded 0.8, suggesting the scale's reliability (Table 2).

5.2. Pilot study

A pilot study was conducted to test for manipulation checks. This experiment used a 2 × 2 factorial design with ethical endowments: embedded in the AI build versus not embedded, and delegation levels: human selection of decision versus autonomous control. The setup assumes that (1) respondents can recognize the two levels of ethical endowment and (2) they can evaluate and distinguish the levels of delegation. Respondents were randomly assigned to all four situations to adjust for ordering effects using Qualtrics' randomization function. The pilot study focused on understanding manipulation checks through a within-subjects design to discern cognitive differences among various choices. Within-subjects designs effectively minimize variability arising from individual differences, which is essential for assessing the subtle effects of manipulation checks (Charness et al., 2012). Furthermore, before conducting a full-scale study, within-subjects designs are optimal for exploratory phases such as pilot testing to evaluate the feasibility and clarity of experimental manipulations (Greenwald, 1976).

A total of 115 business college graduate students (53.91 % male) at a prominent southwestern institution in the United States were provided the scenarios used in Experiment 1 (as shown in Appendix B) and were asked two questions after each scenario: 1) In this scenario, is the agentic IS ethical? and 2) In this scenario, does the hiring manager have the autonomy to make the decision? The overall within-subjects ANOVA for both ethics embedded (F=69.653, $p < 0.001$) and autonomy (F=15.267, $p < 0.001$) was significant. The mean difference between ethics-embedded and not-embedded ethics was 1.75 ($p < 0.001$), whereas the mean difference between human decision-selection and autonomous control was 1.10 ($p < 0.001$) and 1.22 ($p < 0.001$). Based on the pilot results, we revised the verbiage of the experiment scenarios.

5.3. Experiment 1

5.3.1. Participants and the procedure

We recruited 209 respondents on Prolific.co, a research data collection platform based in the United Kingdom. There were three recruitment requirements: 1) respondents must be between 18 and 60 years old, 2) they must have a full-time or part-time job, and 3) they must use information systems daily in their jobs. In addition, the distribution of respondents was balanced based on gender.

Prolific was selected as our data collection platform due to its advantages over other crowdsourcing platforms. Prolific provides access to a diverse, international participant pool with proven reliability in producing high-quality data for academic research (Palan & Schitter, 2018). The platform features built-in attention check capabilities and pre-screening options that allowed us to target participants who regularly use information systems in their professional roles, enhancing the relevance of our sample to the research context. Additionally, Prolific offers transparent participant compensation policies and streamlined integration with our survey platform, Qualtrics, facilitating a smooth data collection process.

The Prolific project was linked with a Qualtrics survey, which used randomization features to assign respondents to a specific scenario in the 2 × 2. The Qualtrics survey had skip logic built for consent and the three aforementioned requirements. Moreover, the Qualtrics survey had two attention check questions, one in the middle and one near the end, to ensure the attentiveness of the respondents. The instrument was tested for randomization and average completion time before deployment.

All submissions were manually reviewed for completeness. A total of 220 responses were collected, of which 11 were rejected due to incompleteness. The completed responses included 103 females, 103 males, and three who identified as others. The average time for completing the experiment was 8 min and 2 s. Table 3 shows the respondents' country of residence.

5.3.2. Experiment Design

Experiment 1 used a 2 × 2 factorial design to manipulate ethics-driven endowment and delegation level. Ethics-driven endowment has two levels: ethics endowed versus not endowed in the agentic IS build. The scenarios incorporate these by discussing the agentic IS data and its acquisition methods. In the context where ethical endowment, the agentic IS, HireIQ, "scrapes the applicants' web and social media feeds and uses third-party websites to obtain additional data about the applicants without their knowledge or consent." HireIQ uses this data in its

Table 2
Construct reliability, convergent validity, discriminant validity, and correlations.

	EA	DE	TE	EJ	Cronbach's alpha	CR	AVE
EA	**(0.934)**				0.954	0.953	0.872
DE	0.765	**(0.778)**			0.873	0.858	0.604
TE	0.40	0.59	**(0.741)**		0.831	0.814	0.531
EJ	0.738	0.86	0.685	**(0.789)**	0.872	0.867	0.622

EA = Agentic IS Ethical appraisal; DE = Deontological evaluation of delegation; EJ = Ethical judgment; TE = Teleological evaluation of delegation. The lower diagonal elements are correlations (ϕ) estimates between latent constructs, and the diagonal elements (in bold) are the extracted square root of the average variance extracted (AVE).

Table 3
Respondents' country of residence.

Country	Frequency	Percentage
United Kingdom	74	35.41
South Africa	23	11
Poland	16	7.65
Portugal	13	6.22
Germany	11	5.26
Italy	11	5.26
Canada	8	3.82
Greece	8	3.82
Netherland	7	3.35
France	6	2.87
Hungary	6	2.87
Mexico	5	2.39
Spain	5	2.39
Chile	3	1.43
Latvia	3	1.43
United States	3	1.43
Czech Republic	2	0.95
Estonia	2	0.95
Austria	1	0.5
Israel	1	0.5
Switzerland	1	0.5
Total	209	100

hiring algorithms." The second variable manipulated in the experiment is the delegation level. We used two levels from the scale: human selection of decision–agentic IS makes recommendations to the human; the human makes the decision based on the recommendations, and autonomous control–agentic IS makes the decision. In the scenario using autonomous control, "HireIQ automatically flagged and removed several candidates from the search because it found what it considers inappropriate web or social media activity. Following the removal of flagged candidates, based on the application credentials, HireIQ autonomously selected the best candidate for the job." The scenarios represent a hiring manager as the organizational human agent or delegator who delegated the recruitment task to HireIQ. The respondents were asked to put themselves in the job applicant role, a stakeholder directly impacted by the organization's decision to use HireIQ, to evaluate the ethicality of the agentic IS and delegation.

Four dependent variables were measured: agentic IS ethical appraisal, deontological appraisal of delegation, teleological appraisal of delegation, and ethical judgment of delegation. A between-subjects design was employed, in which participants were randomly assigned to different scenarios using Qualtrics' randomization feature. Before the experiment, the instrument was tested to verify the effectiveness of the randomization process. Appendix B shows the scenarios.

5.3.3. Common method bias

Self-reported data from one source can lead to common method bias (CMB). When designing our survey instrument, we kept questions simple, focused, and concise, avoided double-barreled questions and conceptual dependence between dependent and independent variables, used randomized scenarios, and reiterated respondent anonymity and our study's exclusive research purpose (Podsakoff et al., 2003). We employed the marker variable technique recommended by Lindell and Whitney (2001) to assess common method bias. We chose blue attitude (i.e., an individual's attitude toward the color blue) (Miller & Simmering, 2023) as the marker variable because it is not expected to have any bearing on the other variables (Simmering et al., 2015). The low correlations between this marker variable and our study constructs (highest correlation = 0.177) suggest that common method bias was not a significant concern in our dataset. Our VIF analysis further supported this conclusion, where all values were below the 3.33 threshold suggested by Kock (2015).

5.3.4. Results

5.3.4.1. Direct effect of ethics-driven endowment on agentic IS ethical appraisal. We conducted a 2 (ethical endowment: embedded in the AI build versus not embedded) x 2 (level of delegation: autonomous versus human selection of recommendation) analysis of variance (ANOVA) on the agentic IS ethical appraisal. The results, as shown in Fig. 3, indicated that the participants perceived the agentic IS to be more ethical if the ethical endowments are embedded ($M = 3.6085$, $SD = 1.0690$) than the agentic IS that does not embed ethical endowments ($M = 2.0291$, $SD = 1.1372$), and that this difference was statistically significant ($F(1, 209) = 106.228$, $p < 0.0001$, partial $\eta^2 = 0.34$). The results show a large effect size (partial $\eta^2 = 0.34$) per Cohen's (1988) guidelines. On the contrary, the main effect of level of delegation was not significant ($F(1, 209) = 0.001$, $p = 0.980$, partial $\eta^2 < 0.01$) on the agentic IS ethical appraisal, which supports the model. In addition, the interaction effect was not significant ($F(1, 209) = 0.279$, $p = 0.598$). Table 4 shows the results of the between-subjects ANOVA. The results support H1.

5.3.4.2. The direct effect of the level of delegation on the deontological evaluation of delegation. We performed another ANOVA on the deontological evaluation of delegation. Fig. 4 shows that the respondents perceived agentic IS with ethical endowments to be more deontologically ethical ($M = 3.1840$, $SD = 0.8173$) than when ethical endowments were not embedded in the build of agentic IS ($M = 2.3252$, $SD = 0.9577$). This difference was significant ($F(1, 209) = 49.273$, $p < 0.001$). However, the main effect of the level of delegation was not significant ($F(1, 209) = 2.067$, $p = 0.152$), providing no support for H3. The interaction effect was also not significant ($F(1, 209) = 1.648$, $p = 0.201$). Table 5 shows the results of the between-subjects ANOVA.

5.3.4.3. The direct effect of the level of delegation on teleological evaluation of delegation. In another ANOVA conducted on teleological evaluation of delegation, the results, as shown in Fig. 5, indicated that the respondents perceived agentic IS with ethical endowments to be more teleologically ethical ($M = 3.2524$, $SD = 0.84268$) than agentic IS with no ethical endowment ($M = 2.911$, $SD = 0.814$), and this difference was significant ($F(1, 209) = 8.990$, $p = 0.0003$). However, the main effect of the level of delegation was insignificant ($F(1, 209) = 1.354$, $p = 0.246$), providing no support for H4. The interaction effect was significant at $\alpha = 0.05$ ($F(1, 209) = 4.125$, $p = 0.044$). Table 6 shows the results of the between-subjects ANOVA.

5.3.4.4. Structural model assessment and results. The established research hypotheses were collectively tested using R Lavaan CB-SEM. To achieve this objective, we substituted the covariance between constructs in the previous CFA model with the causal associations illustrated in Fig. 2.

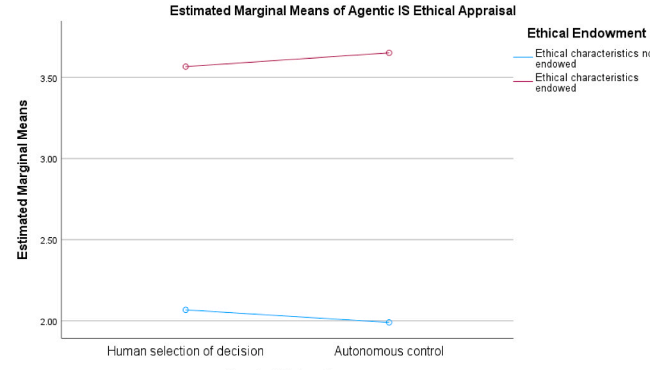

Fig. 3. Effect of ethics-driven endowment on agentic IS ethical appraisal.

Table 4
Between-subjects ANOVA on agentic IS ethical appraisal.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	130.649[a]	3	43.550	35.488	< .001
Intercept	1660.005	1	1660.005	1352.705	< .001
EE	130.361	1	130.361	106.228	< .001
LOD	.001	1	.001	.001	.980
EE * LOD	.343	1	.343	.279	.598
Error	251.571	205	1.227		
Total	2056.250	209			
Corrected Total	382.220	208			

a. R Squared = .342 (adjusted R Squared =.332)
EE = Ethics-driven endowment; LOD = Level of delegation

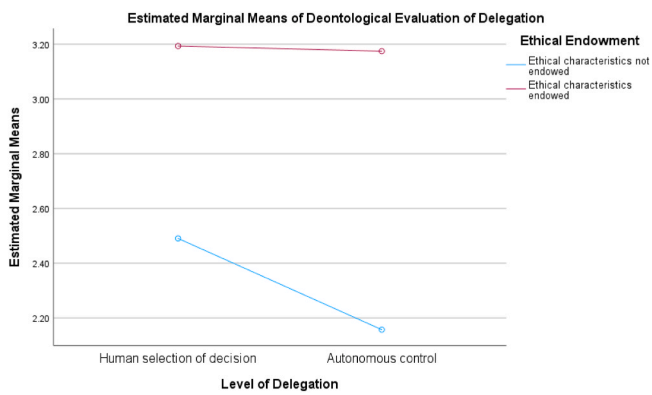

Fig. 4. Effect of level of delegation on the deontological evaluation of delegation.

Table 5
Between-subjects ANOVA on the deontological evaluation of delegation.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	41.395[a]	3	13.798	17.584	< .001
Intercept	1584.523	1	1584.523	2019.244	< .001
EE	38.665	1	38.665	49.273	< .001
LOD	1.622	1	1.622	2.067	.152
EE * LOD	1.293	1	1.293	1.648	.201
Error	160.866	205	.785		
Total	1795.222	209			
Corrected Total	202.260	208			

a. R Squared = .205 (adjusted R Squared =.193)
EE = Ethics-driven endowment; LOD = Level of delegation

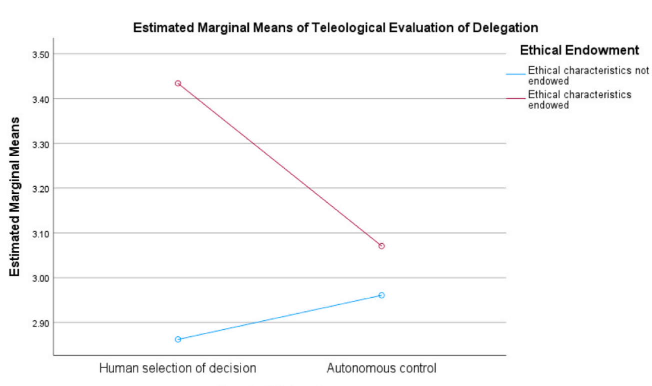

Fig. 5. Effect of the level of delegation on teleological evaluation of delegation.

Table 6
Between-subjects ANOVA on teleological evaluation of delegation.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	9.833[a]	3	3.278	4.855	.003
Intercept	1984.627	1	1984.627	2939.6	< .001
EE	6.070	1	6.070	8.990	.003
LOD	.914	1	.914	1.354	.246
EE * LOD	2.785	1	2.785	4.125	.044
Error	138.403	205	.675		
Total	2136.215	209			
Corrected Total	148.236	208			

a. R Squared = .605 (Root MSE =.723)
EE = Ethics-driven endowment; LOD = Level of delegation

The hypothesized research model exhibited a good fit with the observed data (χ^2/df =1.896, NFI = 0.918, NNFI = 0.951, CFI = 0.959, TLI = 0.951, RMSEA = 0.06, and SRMR = 0.054). CFI and TLI values above 0.95, RMSEA below 0.06, and SRMR below 0.08 generally indicate a good fit (Hu & Bentler, 1999). The item loadings in the measurement model were similar to those in the CFA. Regarding nomological validity, the path estimates and variance explained (R^2 value) for most of the dependent variables in the structural model were of greater significance. The model explained 38 % of the variance in agentic IS ethical appraisal, 64.4 % of the variance in deontological evaluation of delegation, and explained 83.7 % of the variance in ethical judgment of delegation.

Ethics-driven endowment was a significant predictor of agentic IS ethical appraisal (β = 0.616, $p < 0.001$), supporting H1. Similarly, agentic IS ethical appraisal was found to be a significant predictor of deontological evaluation of delegation (β = 0.803, $p < 0.001$), thereby supporting H2. However, the level of delegation was not found to be a significant predictor of deontological and teleological evaluation of delegation. Deontological evaluation of delegation was found to be a strong predictor of ethical judgment of delegation (β = 0.915, $p < 0.001$), supporting H5. Similarly, teleological evaluation of delegation was found to be a strong predictor of ethical judgment of delegation (β = 0.683, $p < 0.001$), supporting H6. Table 7 and Fig. 6 show the CB-SEM analysis results.

The non-significant effects of the level of delegation on deontological evaluation (H3; $\beta = -0.008$, $p = 0.933$) and teleological evaluation (H4; $\beta = -0.06$, $p = 0.565$) in Experiment 1 warrant further investigation. The absence of a significant effect for the level of delegation manipulation in the 2 × 2 factorial experiment can be attributed to the reduced statistical power inherent in complex designs and the potential masking

Table 7
CB-SEM results of the hypothesized relationships.

Hypothesized Relationships	Path Coefficients	z-stats	Supported?
H1: Ethics-driven endowment → Agentic IS Ethical appraisal	0.616	10.331	Yes***
H2: Agentic IS Ethical appraisal → Deontological evaluation of delegation	0.803	12.889	Yes***
H3: Level of delegation→ Deontological evaluation of delegation	−0.008	−0.084	Not supported
H4: Level of delegation → Teleological evaluation of delegation	−0.567	−0.060	Not supported
H5: Deontological evaluation of delegation → Ethical judgment of delegation	0.915	14.205	Yes***
H6: Teleological evaluation of delegation → Ethical judgment of delegation	0.683	6.823	Yes***

Significance: * 0.05, ** 0.01, *** 0.001

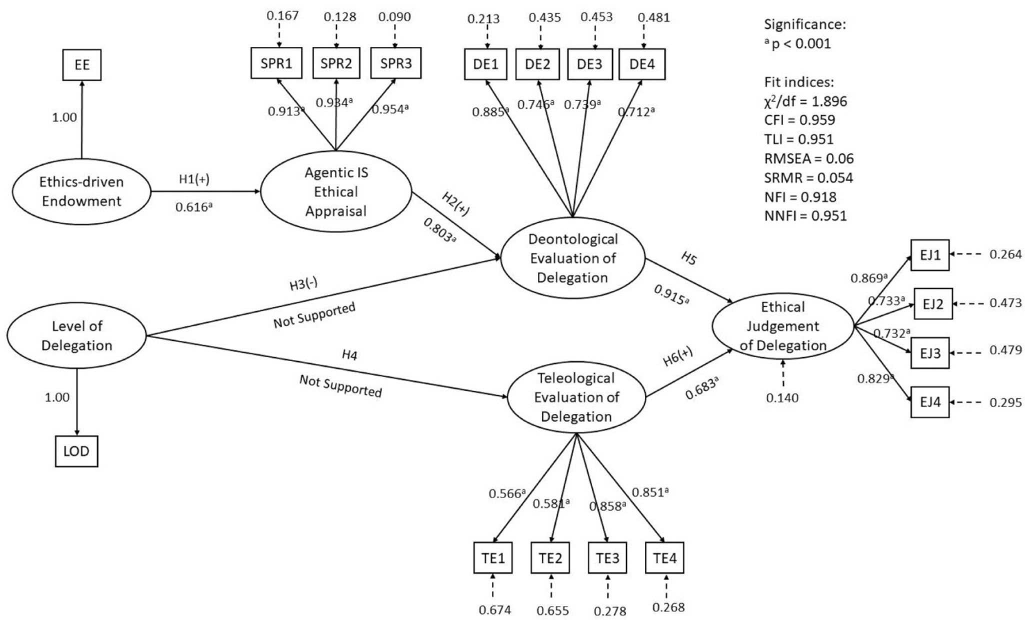

Fig. 6. CB-SEM model with fit indices.

or attenuation of its main effect by interactions or contextual factors. Allocating participants across multiple cells reduces the sample size per condition in factorial designs, thereby diminishing the power to detect main effects of smaller magnitude (Cohen, 1988; McClelland, 1997). Moreover, interactions between manipulated factors can mask or attenuate the main effects. When two variables are tested simultaneously, one manipulation may alter the context in which the other operates, resulting in conditional or context-dependent effects that do not emerge as significant main effects. Finally, if the manipulation strength or reliability checks are not equally robust across all conditions, complex designs increase the risk of Type II errors (Maxwell et al., 2017).

To address this methodological limitation related to the sensitivity and interpretability of main effects under complex experimental conditions, we performed a follow-up experiment that isolates the level of delegation, eliminating potential interference from ethical endowments and maximizing power for detecting the level of delegation's effect (Maxwell et al., 2017). Experiment 2 used a within-subjects design that allowed participants to compare different levels of delegation.

5.4. *Experiment 2*

To isolate the level of delegation while minimizing inferences from ethical considerations, Experiment 2 focused solely on the level of delegation. Using a within-subjects design, participants experienced all scenarios to assess their preferences across five levels of delegation adapted from Vagia et al. (2016). The scenario sequence was randomized using Qualtrics, and we ensured that the randomization worked as intended.

For several compelling reasons, a within-subjects design was adopted for Experiment 2. First, by controlling for individual differences, this approach substantially increases statistical power, allowing us to detect smaller effect sizes with the same sample size compared to between-subjects designs (Charness et al., 2012). Second, within-subjects designs better reflect real-world decision contexts in which individuals evaluate multiple AI delegation options sequentially or comparatively, enhancing ecological validity. Third, this design enables us to examine how the same individuals evaluate different levels of delegation, revealing nuanced preference patterns that might be obscured in between-subjects comparisons (Charness et al., 2012; Greenwald, 1976). While we acknowledge potential carryover effects, we mitigated them by randomizing the order of scenario presentation.

5.4.1. *Participants and the procedure*

Experiment 2 was also conducted on Prolific and used the same exact requirements as Experiment 1. However, the respondents from Experiment 1 were not allowed to participate in Experiment 2. The distribution of respondents was balanced based on gender. The Prolific project used Qualtrics survey randomization to randomize scenario sequences. The instrument included two attention checks, one in the middle and one at the end, to ensure respondents' attention. The instrument was tested for randomization and average completion time before deployment. All submissions were manually reviewed for completeness. A total of 315 responses were collected, of which 16 were rejected due to incompleteness. The completed responses were from 150 women and 149 men. The median completion time was 14 min and 22 s. Table 8 shows

Table 8
Respondents' country of residence.

Country	Frequency	Percentage
United Kingdom	135	45.2
Poland	28	9.4
Mexico	25	8.4
Portugal	22	7.4
South Africa	20	6.7
Italy	18	6
Canada	9	3
Spain	7	2.3
Belgium	4	1.3
Germany	4	1.3
Greece	4	1.3
United States	4	1.3
Finland	3	1
Hungary	3	1
Chile	2	0.7
Czech Republic	2	0.7
Estonia	2	0.7
France	2	0.7
Netherlands	2	0.7
Latvia	1	0.3
Slovenia	1	0.3
Switzerland	1	0.3
Total	299	100 %

the respondents' country of residence.

Each participant was asked to evaluate all five scenarios (Appendix B). Similar to Experiment 1, the participants were instructed to assume the position of the job candidate to evaluate the viewpoint of the stakeholders. The scenarios involved the same context, but the level of delegation was manipulated to five levels of delegation, ranging from manual control to autonomous control, describing different levels of AI autonomy. The sequence of scenarios was randomized using Qualtrics' randomization features to control for order effects.

5.4.2. Results

5.4.2.1. Effect of level of delegation on the deontological evaluation of delegation. The results of the within-subjects ANOVA on the deontological evaluation of delegation indicated that participants rated agentic IS with higher levels of autonomy as less ethically acceptable from a deontological perspective. The overall model was found to be significant ($F(1, 1494) = 570.67$, $p < 0.0001$, partial $\eta^2 = 0.323$), suggesting that the deontological evaluation of delegation is statistically different across different levels of delegation. The results show a large effect size (partial $\eta^2 = 0.323$) per Cohen's (1988) guidelines. Post-hoc Tukey analysis reveals four subsets, suggesting that manual control (M = 3.5811, SD = 0.784) was statistically different from decision proposal (M = 3.3746, SD = 0.7105), human selection of decision (M = 3.182, SD = 0.8077), and computer selection of decision (M = 2.6513, SD = 0.7937). However, the computer selection of decision and autonomous control were not statistically different. The results of our analysis are shown in Tables 9 and 10. Fig. 7 plots the deontological evaluation of delegation against the level of delegation. The results strongly support H3.

The 'Sig.' values in Table 10 represent the p-values for the homogeneous subsets identified through Tukey's honestly significant difference post-hoc test. Values close to 1.000 indicate high homogeneity within the subset, confirming that the means within each subset are not significantly different. The different subsets (columns 1–4) represent groupings whose means differ significantly. The significance value of 0.542 for Subset 1 suggests that computer selection of decision and autonomous control do not exhibit a statistically significant difference ($p > 0.05$). Both levels significantly differ from the other three delegation levels, constituting distinct subsets.

5.4.2.2. Effect of the level of delegation on teleological evaluation of delegation. The results of the within-subjects ANOVA on teleological evaluation of delegation indicated that participants rated agentic IS with higher levels of autonomy as less ethically acceptable from a teleological perspective. However, the pattern is different from that of the deontological evaluation. The overall model was found to be significant ($F(1, 1493) = 224.16$, $p < 0.0001$, partial $\eta^2 = 0.057$), suggesting that teleological evaluation of delegation is statistically different across different levels of delegation. The results show a medium effect size (partial $\eta^2 = 0.057$) per Cohen's (1988) guidelines. Post-hoc Tukey analysis reveals two subsets, suggesting that the manual control (M = 3.3174, SD = 0.057) was the most effective. = 0.789) was not statistically different from the decision proposal (M = 3.3319, SD = 0.745) and human selection of decision (M = 3.3456, SD = 0.8903). However, computer selection of decision and autonomous control were statistically different from the first subset. The results of our analysis are shown in Tables 11 and 12. Fig. 8 plots the teleological evaluation of delegation against the level of delegation. The results strongly support H4.

The significant findings in Experiment 2, which used a within-subjects design that allowed for direct comparison of delegation levels, support this interpretation and suggest that the comparative context is crucial for eliciting ethical judgments related to delegation.

The findings from Experiment 2 provide strong support for H3 and H4, demonstrating that from both deontological and teleological perspectives, users consistently evaluate higher levels of AI autonomy as less ethically acceptable. The evaluation pattern differs slightly between the two ethical frameworks: from a deontological perspective, each increase in autonomy (except between computer selection and autonomous control) resulted in significantly lower ethical evaluations, whereas the key distinction was between systems with and without human oversight. These findings indicate that human involvement in AI decision processes is crucial for user acceptance.

Beyond statistical significance, we examined effect sizes to assess the practical significance of our findings. For Experiment 1, the effect of ethics-driven endowments on agentic IS ethical appraisal yielded a large effect size (partial $\eta^2 = 0.34$), indicating that the presence of ethics-driven endowments explained approximately 34 % of the variance in ethical appraisal. This represents a substantial effect according to

Table 10
Mean of groups in homogeneous subsets using Tukey[a].

Level of delegation	N	Subset 1	2	3	4
Computer Selection of Decision	299	2.6513			
Autonomous Control	299	2.7494			
Human Selection of Decision	299		3.182		
Decision Proposal	299			3.3746	
Manual Control	299				3.5811
Sig.		0.542	1.000	1.000	1.000

a. Uses harmonic mean sample size = 299.00

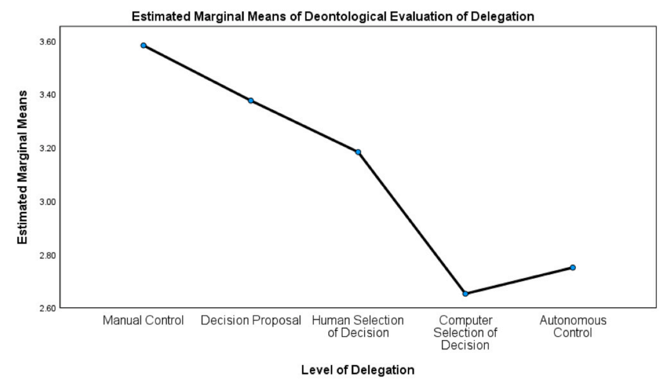

Fig. 7. Estimated marginal means for the deontological evaluation of delegation.

Table 9
Within-subjects ANOVA on the deontological evaluation of delegation.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Model	747.891[a]	299	2.501	8.38	< .0001
Id	577.589	298	1.938	6.49	< .0001
LOD	170.301	1	170.301	570.67	< .0001
Error	356.618	1195	0.298		
Corrected Total	1104.509	1494			

a. R Squared = .677 (Root MSE =.5463); LOD = Level of delegation

Table 11
Within-subjects ANOVA on teleological evaluation of delegation.

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Model	776.4539[a]	299	2.5968	9.48	< .0001
Id	715.069	298	2.399	8.76	< .0001
LOD	61.3838	1	61.3848	224.16	< .0001
Error	1016.995	1194	0.2738		
Corrected Total	1103.419	1493			

a. R Squared = .7037 (Root MSE =.523); LOD = Level of delegation

Table 12
Mean of groups in homogeneous subsets using Tukey[a].

Level of delegation	N	Subset	
		1	2
Computer Selection of Decision	299	2.8311	
Autonomous Control	299	2.8514	
Manual Control	299		3.3174
Decision Proposal	299		3.3319
Human Selection of Decision	298		3.3456
Sig.		0.998	0.994

a. Uses harmonic mean sample size 298.799

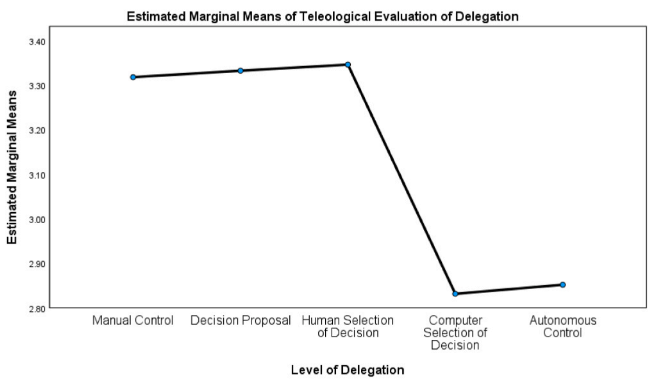

Fig. 8. Estimated marginal means for teleological evaluation of delegation.

Cohen's (1988) guidelines. In contrast, the level of delegation showed a negligible effect size (partial $\eta^2 < 0.01$). In Experiment 2, the level of delegation had a statistically significant effect on deontological evaluation ($F(1, 1195) = 570.67$, $p < 0.0001$, partial $\eta^2 = 0.323$), representing a large effect size. For teleological evaluation, the level of delegation also had a significant effect ($F(1, 1194) = 224.16$, $p < 0.0001$, partial $\eta^2 = 0.057$), representing a medium effect size.

6. Discussion

This study explores stakeholders' ethical judgment of delegation to agentic IS. In particular, we aimed to answer the following three questions: How does the ethical judgment of stakeholders regarding agentic IS change when these systems incorporate ethical endowments? Does the level of delegation impact the ethical judgment of delegation to agentic IS? To what degree do stakeholders prioritize goal orientation or ethical considerations when utilizing agentic IS for delegation?

The results of Experiment 1 indicate that stakeholders are more likely to perceive agentic IS as ethical when they possess ethical endowments. The findings indicate that ethics-driven endowments are essential for a more favorable ethical assessment of agentic IS. The results also emphasize the importance of signaling the presence of ethical endowments to stakeholders, who may be unaware of them if they are not explicitly signaled.

The literature on Explainable AI (XAI) highlights the importance of signaling the embodiment of ethical and trustworthy features in AI systems that align utilitarian performance with the rightness expectations of stakeholders. Several studies have documented how concrete XAI signaling mechanisms have tangibly influenced user perceptions and decision-making in real-world settings. For example, Lundberg and Lee's (2017) SHAP framework was incorporated into a clinical decision-support system for predicting sepsis. This system used color-coded force plots to dynamically illustrate the positive or negative contribution of each biomarker to the overall risk scores. In user evaluations, physicians noted that having access to these individualized explanations increased their confidence in AI recommendations and encouraged them to critically assess any unusual predictions. This practice promoted transparency and demonstrated a commitment to patient safety (Lundberg & Lee, 2017). Similarly, Ribeiro et al. (2016) local interpretable model explanations were used in Fair Isaac Corporation's credit-scoring platform to emphasize feature importance at the individual applicant level. Consumers who saw local surrogate explanations for declined applications were more likely to trust the process and identify and correct data errors, demonstrating interpretability as an ethical signal of fairness and recourse (Ribeiro et al., 2016).

Our findings regarding the positive influence of ethics-driven endowments on agentic IS ethical appraisal can be interpreted through signaling theory (Spence, 1978). Ethics-driven endowments function as quality signals that reduce information asymmetry between the agentic IS and its stakeholders. When organizations explicitly implement and communicate ethical endowments in their agentic IS, ethical commitment and reliability are signaled, addressing stakeholders' uncertainty about potential ethical risks. This signaling mechanism creates a trust foundation (Doshi-Velez & Kim, 2017; Lundberg & Lee, 2017; Ribeiro et al., 2016) that enhances the ethical appraisal of the agentic IS and the subsequent ethical evaluation of delegation.

Without explicit signals on an AI's ethical design or safeguards, stakeholders depend on established schemas and heuristics. Studies indicate that autonomous systems, particularly those regarded as agentic, frequently elicit inherent caution and skepticism stemming from apprehensions over control, predictability, and moral agency (Burton et al., 2020; Shin, 2021). In such instances, the preconceived notions that agentic IS are deontologically unethical may result in a reduced ethical evaluation of the system, the delegation using the system, and ultimately the intent to use the system.

The findings from Experiment 2 indicate that stakeholders evaluate the ethicality of different levels of delegation differently. Stakeholders perceive autonomous decisions with minimal human involvement as less ethical than those made by agentic information systems that keep humans involved. The consensus is that a higher degree of AI autonomy is perceived as less ethically sound, leading to a preference for human oversight in making final decisions with AI. The findings also indicate that stakeholders using agentic IS exhibit mixed deontological-teleological principles. According to H-V theory, adherence to exclusively deontological or teleological ethics is improbable among individuals in various contexts. Consequently, the mixed deontological-teleological perspective should be regarded as a continuum, in which specific individuals are more attuned to the utility of agentic IS, while others prioritize ethical considerations.

The preference for human oversight in AI decision-making processes revealed in Experiment 2 connects to accountability theory (Lerner & Tetlock, 1999). Stakeholders' lower ethical evaluations of highly autonomous agentic IS likely stem from concerns about diffused responsibility and reduced accountability. Human oversight creates a clear line of responsibility, maintaining what Nissenbaum (1996) calls the "moral agency gap"—the concern that fully autonomous systems cannot be held morally responsible for their actions like humans can (Martin, 2019). This accountability concern appears to influence both deontological (based on the principle that decision authority should remain with moral agents) and teleological (based on the anticipated consequences of removing human oversight) evaluations.

In sum, this study contributes to the ongoing discourse in information systems research on ethical human–AI collaboration by demonstrating that the ethical endowments embedded in agentic IS shape the stakeholders' evaluations of the agentic IS, the delegation to the agentic IS, and how those capabilities are signaled and contextualized. The findings also underscore that delegation's perceived ethicality varies with the level of autonomy, reflecting stakeholders' preference for human oversight in agentic IS delegations. Finally, the findings indicate that stakeholders follow a mixed deontological-teleological system for agentic IS delegations, balancing goal orientation with ethical considerations.

7. Implications

This study contributes to the literature on AI ethics and user acceptance. First, we incorporate ethical constraints into the agentic IS delegation framework, addressing the need to address ethical concerns pertaining to agentic IS. Neglecting ethical considerations can lead to AI initiative failures (Pethig & Kroenung, 2022), negative publicity, and substantial waste of resources (Crockett et al., 2021; Rinta-Kahila et al., 2021). This study enhances its robustness and practical applicability of the agentic IS delegation framework by incorporating ethics into it. Moreover, we broaden the notion of agency in agentic IS delegation by employing stakeholder theory, which facilitates the inclusion of essential stakeholders, such as consumers, users, and customers, within the agentic IS delegation framework. Building upon the foundational work of Baird and Maruping (2021), this study provides a comprehensive operationalization of key constructs and establishes directional relationships within the model. This advancement enables rigorous empirical testing across diverse contexts, facilitating a more nuanced understanding of agentic IS.

Second, the study introduces a novel conceptualization of agentic IS endowments, distinguishing between performance- and ethics-driven endowments. This construct contributes to the literature on value-sensitive design (Friedman et al., 2013) by operationalizing how ethical endowments can be systematically embedded in technological artifacts and demonstrating that these endowments significantly influence ethical evaluations. Moreover, the novel construct can be operationalized with different ethical endowments such as fairness, bias, discrimination, privacy, transparency, and so on, enabling researchers to evaluate diverse ethical issues across different scenarios and circumstances.

Third, our investigation of levels of delegation contributes to the IS literature by elucidating how varying degrees of delegation in human-AI interactions impact ethical perceptions. The proposed delegation levels add a new, nonlinear dynamic to delegation theory, likely driven by a perceived accountability gap. Expanding on our theoretical discourse regarding the impact of severity of outcomes on ethical assessments in AI delegation (as delineated in the formulation of H4 and H5), we advocate for a risk-based paradigm for human-in-the-loop design as a practical implication for management practice. The degree of human oversight should be customized according to the ethical and operational significance of the task. Preserving human oversight is essential for establishing ethical accountability and sustaining stakeholder trust in high-stakes decisions—such as hiring, credit approvals, or disciplinary actions—where mistakes have substantial moral and reputational repercussions. In contrast, increased automation can be safely adopted to improve operational efficiency in low-risk, routine tasks (e.g., scheduling or basic data processing), where the potential of harm is negligible. This adaptive strategy correlates ethical accountability with risk exposure, providing a practical framework for AI integration in organizational settings.

Lastly, instrumental stakeholder theory asserts that actively addressing stakeholder interests is essential for achieving an enduring competitive advantage (Jones et al., 2018). When applied to agentic IS, this perspective reconceptualizes the dual imperatives of efficiency and ethics as a strategic complement rather than a trade-off. Companies fulfill normative stakeholder expectations and foster the trust and reputational capital essential for sustained user adoption and market performance by integrating and transparently communicating ethical attributes such as fairness, privacy protection, and accountability (Jones et al., 2018). Consequently, designing ethical AI systems and proactively communicating their moral safeguards emerges as a fundamental business strategy—utilizing ethical signaling to diminish information asymmetry, alleviate stakeholder skepticism, and secure sustainable competitive advantages, rather than functioning solely as a compliance measure.

8. Limitations and future research

This study has several limitations that should be acknowledged. First, our operationalization of ethics-driven endowments focuses on privacy protections, representing one dimension of ethical AI. Other ethical considerations, such as fairness, transparency, and accountability, may influence ethical evaluations differently and warrant separate investigation. Second, while offering strong internal validity, the scenario-based experimental design may not fully capture the complexity of real-world AI implementation contexts. The participants' responses to hypothetical scenarios might differ from their reactions to actual agentic IS encountered in organizational settings. Third, although our sample was diverse internationally, it was recruited through Prolific, which may introduce self-selection biases. Participants on crowdsourcing platforms may have characteristics that differ from those of the general population or specific organizational stakeholders who would evaluate agentic IS in practice. Finally, we encountered a high measurement overlap, or lack of discriminant validity, between the deontological evaluation of delegation and the ethical judgement of delegation. For future research, researchers can develop measures that clearly distinguish between the two constructs or explore evaluating the model with pluralistic ethical scales, such as the one developed by Reidenbach and Robin (1988).

Several promising directions for future research emerged from this study. First, researchers can utilize the TEAM to expand the investigation to other ethics-driven endowments beyond privacy, such as fairness-aware algorithms, explainability features, and bias mitigation techniques. This provides a more comprehensive understanding of how different ethical dimensions influence user evaluations. Second, field studies examining actual implementations of agentic IS with varying levels of delegation and ethical features would complement our experimental findings and enhance ecological validity. Case studies of organizations that successfully balance efficiency and ethical considerations could provide valuable practical insights. Third, exploring potential moderators of the relationships identified in our model—such as domain criticality, user expertise, cultural factors, and individual ethical orientations—would help identify boundary conditions and contextual influences on ethical evaluations of AI. Fourth, in our scenarios, the ethical framing was consistently presented before the level of delegation. Future research can systematically manipulate the sequence of these two variables to examine potential framing or primacy effects. Such an investigation will elucidate the extent to which information sequence influences the relationship between delegation level and ethical endowments, providing practical insights for designing AI user interfaces. Finally, the model can be augmented to evaluate task difficulty and the participation of various stakeholders in scenario-based or action research.

9. Conclusion

The adoption and ethical implementation of agentic IS present organizations with complex challenges at the intersection of technological capability, ethical responsibility, and user acceptance. Our research makes three significant contributions to addressing these challenges. First, we demonstrate that stakeholders employ different ethical frameworks when evaluating agentic IS, valuing goal orientation and ethical considerations through a mixed deontological-teleological approach. Second, we show that ethics-driven endowments significantly enhance ethical evaluations of both agentic IS and the delegation to agentic IS, highlighting the importance of embedding ethical considerations during agentic IS design and signaling the presence of ethical endowments to stakeholders. Third, a consistent preference for human oversight in AI decision-making processes is revealed, challenging the assumption that full automation is always preferable.

These findings have important implications for organizations implementing agentic IS. Organizations can enhance user acceptance by designing agentic systems with embedded ethical endowments, maintaining appropriate levels of human oversight, and addressing both performance and ethical dimensions in communication while realizing the efficiency benefits of agentic IS. The TEAM model developed in this study provides a structured framework for balancing and testing these considerations.

As AI technologies evolve and permeate organizational processes, attending to stakeholders' ethical evaluations is essential for responsible innovation. Organizations that thoughtfully design agentic IS with ethics-driven endowments and appropriate levels of human oversight will align with stakeholders' ethical expectations and potentially accelerate AI adoption. Ultimately, the sustainable advancement of agentic IS depends on our ability to balance technological innovation with ethical reflection, creating systems that enhance human capabilities while respecting fundamental ethical principles.

Declarations

The author(s) disclose that there are no direct or indirect conflicts of interest within or outside this timeframe. The authors have not received any grants from funding agencies and/or research support (including salaries, equipment, supplies, reimbursement for attending symposia, and other expenses) from organizations that may gain or lose financially through the publication of this manuscript. Moreover, the author(s) have no past, recent, or future employment offer(s) from organizations that may gain or lose financially through the publication of this manuscript. Additionally, the author(s) (including holdings of their spouses and children) have no financial or non-financial interests from an organization and/or individual that may gain or lose financially or otherwise from the publication of this manuscript.

Compliance with ethical standards

i. **Disclosure of potential conflict of interest**
   The author(s) disclose that there are no direct or indirect conflicts of interest within the three years of beginning the work or outside of this timeframe. The authors have not received any grants from funding agencies and/or research support (including salaries, equipment, supplies, reimbursement for attending symposia, and other expenses) by organizations that may gain or lose financially through publication of this manuscript. Moreover, the author(s) have no past, recent, or future employment offer(s) form organizations that may gain or lose financially through the publication of this manuscript. Additionally, the author(s) (including holdings of their spouses and children) have no financial or non-financial interests from an organization and/or individual that may gain or lose financially or otherwise from the publication of this manuscript.
ii. **Research involving Human Participants and/or Animals**
   The study involved human subjects. However, IRB approval from University of North Texas IRB Review Board was obtained prior to data collection. The approved IRB number is IRB-22–397.
iii. **Informed consent**

Participants were provided the consent information before the data collection using Qualtrics. The survey included break logic, which took the participants to the end of the survey (without data collection) if the consent was not provided. In addition to providing the consent information in plain text, the IRB and the informed consent forms were attached as snapshots in the consent question in Qualtrics.

CRediT authorship contribution statement

Victor R. Prybutok: Writing – review & editing, Validation, Supervision. **Kashif Saeed:** Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijinfomgt.2025.102976.

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Kashif Saeed is an Assistant Professor at the University of North Texas' G. Brint Ryan College of Business. He graduated from N.E.D. University of Engineering with a bachelor's degree in Electrical Engineering, the University of Houston with a master's degree in Computer Science, and the University of North Texas with a doctorate in Information Systems. Kashif has more than 20 years of expertise in information technology, having designed and managed data and analytics solutions for several notable enterprises. His work has appeared in MISQ Executive, Technological Forecasting and Social Change, and the Journal of Computer Information Systems.

Victor R. Prybutok is a G. Brint Ryan Endowed Professor and a Regents Professor of Decision Sciences in the G. Brint Ryan College of Business at the University of North Texas. He received, from Drexel University, a B.S. with High Honors in 1974, a M.S. in 1976, a M. S. in Environmental Health in 1980, and a Ph.D. in Environmental Analysis and Applied Statistics in 1984. Recent awards he received include the 2017 American Society for Quality Gryna Award, the 2018 Decision Sciences Institute Lifetime Distinguished Educator Award, and the 2020 Southwest Decision Sciences Institute Distinguished Service Award. Dr. Prybutok started his service as president elect of the Decision Sciences Institute in May of 2023 and will serve as president starting May of 2024 and past president in 2025.